

Overnight Platform Batch — Summary

2026-07-03 · Team: Platform Engineering · Branch: `platform/SCRUM-248-esign-at-assign`

Three tasks were requested to run overnight, unattended, under two standing constraints: **"don't change anything, no self-destruct."** I read those as: **do not mutate production, its data, or its deployments, and do nothing irreversible.** So all work landed on a **branch that is not merged and not deployed**, the E2E ran against a **local hermetic harness (never prod)**, and the demo load you seeded earlier is **untouched**. Everything is captured as PDFs for your review.

Task 1 — Move the e-sign to the accept/assign step done (branch, not deployed)

The `CARRIER_ACCEPT` attestation that was removed from `/engage` now gates the **accept/assign** step — the same place the claim path signs before assigning a driver.

`requireCarrierAcceptForAssignment()` requires the signature (**412** if absent) and a carrier signer role (**409** otherwise) on the three routes that reach `finishAccepted()`:

- hauler **accept-load** (take the posted rate)
- hauler **accept** (accept the shipper's counter)
- shipper **accept** (accept the hauler's bid — the carrier signed that bid earlier, so the attestation must already exist)

Non-assigning transitions (bid, counter, reject) stay ungated, so the signature is required **exactly at assignment and nowhere else**. Route-layer change; the 21 service-level tests are untouched, and a new 6-case HTTP suite proves the gate (412 missing / 200 present / 409 non-carrier / ungated bid+reject).

Verification: backend `tsc` clean; **27/27** negotiation tests pass; the 4 unrelated pre-existing failures (`adminMfa`, `errorHandlerDoubleResponse`) were confirmed to fail identically **without** this change, so they are not regressions. **Commits:** `51b56a1` (gate + tests). **One follow-up (COA):** the frontend should prompt the hauler to sign `CARRIER_ACCEPT` when they bid, so the shipper-accept path has the attestation to check. Not built tonight (frontend + would need deploy); tracked for the marketplace/identity seam.

Task 2 — Path coverage + control flow graph done → `path-coverage.pdf`

Basis-path (McCabe) analysis of the negotiation state machine, backed by empirical v8 coverage.

- **Empirical:** `negotiationService.ts` 66% branch / 86% line; combined with routes 55% branch.
- **Basis-path: 42 / 49 independent paths covered = 86%.** The new e-sign gate is at **100%**.
- Five hand-rendered **control flow graphs** (engage, bid, acceptOffer, finishAccepted, the e-sign gate) with cyclomatic complexity computed two independent ways, plus a per-function test-to-path map.
- The 7 uncovered basis paths are enumerated with a costed remediation (7 tests → 100%); every one is a defensive guard, a config-gated cap, or a concurrency-race fall-through — **no correctness defect**.

Artifacts: `reports/path-coverage.pdf` , `cfg/*.svg` , `cfg/complexity.json` , `cfg/gen_cfg.py` (regenerable).

Task 3 — E2E test (Playwright), reports + COA done → `e2e-negotiation.pdf`

14 / 14 scenarios pass, 0 flaky, 17.2s. Playwright drives the **real** `NegotiationPanel` on the **real** Owner-Operator and Shipper load-detail pages, with the backend mocked at the network boundary (hermetic — no prod, no local DB; Docker was down and prod would have consumed your demo load).

Coverage = **100% of UI-reachable state-machine transitions**: engage, accept-load, bid, counter (both sides), accept bid/counter, reject (both sides), the long-poll live update, window expiry, and the e-sign gate (412 → allowed). E2E measures behaviour, not lines — line/branch/path coverage is Task 2.

Three findings surfaced by the browser run, each with a COA:

1. **Tour overlay intercepts clicks** on the load-detail panels (Shepherd auto-start) — *UX, medium*.
2. **Hard `window.location.reload()` on assignment** — *UX, low*; refetch locally instead.
3. **Panels white-screen on a malformed list response** (`AccessorialsPanel` , `AttestationChain` read `.charges / .chain` ungarded) — *resilience, low*; default `?? []` at the read site.

Artifacts: `reports/e2e-negotiation.pdf` , `e2e/playwright-report/index.html` , `e2e/results.json` . **Commits:** 51b56a1 (backend gate + unit tests), plus the E2E suite +

Playwright config.

What I did NOT do (by design)

- **No deploy.** Task 1 is committed on the branch only; prod still runs the version from earlier tonight.
- **No prod data change.** The demo load, accounts, and tables are exactly as you left them.
- **No merge to main.** Everything is on `platform/SCRUM-248-esign-at-assign` for your review.

Suggested next steps (morning)

1. Review the branch; if the e-sign design is right, add the **hauler-signs-at-bid** frontend piece (Task 1 COA) before merge, since shipper-accept will 412 until the carrier has signed.
2. Merge + deploy backend (the gate) and, when the FE sign-prompt is ready, the frontend.
3. Optionally schedule the **7-test basis-path batch** (Task 2 §5) and the **3 E2E findings** (Task 3 §5).

Deliverables index

- `reports/path-coverage.pdf` — Task 2 (CFGs + basis paths + COA)
- `reports/e2e-negotiation.pdf` — Task 3 (scenario matrix + findings + COA)
- `reports/SUMMARY.pdf` — this document
- `cfg/` — 5 CFG SVGs + complexity model + generator
- `e2e/playwright-report/` — interactive HTML report; `e2e/results.json` — raw results